(2.1 3) a) $0 \in [0,5), [0,5)$ b) $1 \in (0,5), (0,5], [0,5), [0,5]$

c) $2 \in (0,5), (0,5], [0,5), [0,5], (1,4], [2,3]$

d) $3 \in (0,5), (0,5], [0,5), [0,5], (1,4], [2,3]$

e) $4 \in (0,5), (0,5], [0,5), [0,5], (1,4]$ f) $5 \in (0,5], [0,5]$

(2.1 5) a) nonstop airline flights $\subseteq$ airline flights

b) neither

c) flying squirrels $\subseteq$ living creatures that can fly

(2.1 7) a) $\{1,3,3,3,5,5,5,5\} = \{5,3,1\}$ since order and duplicates don't matter

b) $\{\{1\}\} \neq \{1,\{1\}\}$ since $1 \in \{1,\{1\}\}$ but $1 \notin \{\{1\}\}$

c) $\emptyset \neq \{\emptyset\}$ since $\emptyset \in \{\emptyset\}$ but $\emptyset \notin \emptyset$

(2.1 9) a) $\{x \in \mathbb{R} \mid x$ is an integer greater than $1\}$ yes, $2 \in \mathbb{R}$ s.t. $2 \in \mathbb{Z}$ and $2 > 1$.

b) $\{x \in \mathbb{R} \mid x$ is the square of an integer$\}$ no, $\sqrt{2} \notin \mathbb{Z}$

c) $\{2,\{2\}\}$ yes

d) $\{\{2\},\{\{2\}\}\}$ no, $2 \neq \{2\}$ and $2 \neq \{\{2\}\}$

e) $\{\{2\},\{2,\{2\}\}\}$ no, $2 \neq \{2\}$ and $2 \neq \{2,\{2,\{2\}\}\}$

f) $\{\{\{2\}\}\}$ no, $2 \neq \{\{2\}\}$

(2.1 11) a) $0 \in \emptyset$ false, $\emptyset$ contains no elements

b) $\emptyset \in \{0\}$ false, $\emptyset \neq 0$

c) $\{0\} \subset \emptyset$ false, $\emptyset$ contains no elements

d) $\emptyset \subset \{0\}$ true, $\emptyset$ is a subset of every set

e) $\{0\} \in \{0\}$ false, $\{0\} \neq 0$

f) $\{0\} \subset \{0\}$ false, $\{0\} = \{0\}$

g) $\{\emptyset\} \subseteq \{\emptyset\}$ true, $\emptyset \in \{\emptyset\}$

(2.1 13) a) $x \in \{x\}$ true

b) $\{x\} \subset \{x\}$ false, $\{x\} = \{x\}$

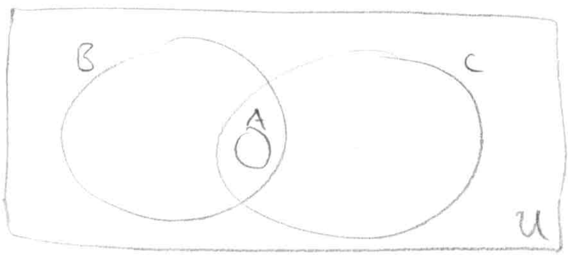
c) $\{x\} \in \{x\}$ false, $x \neq \{x\}$

d) $\{x\} \in \{\{x\}\}$ true

e) $\emptyset \subseteq \{x\}$ true, $\emptyset$ is a subset of every set

f) $\emptyset \in \{x\}$ false, $\emptyset \neq x$

(2.1 18)



(2.1 23) a) $P(\{a\}) = \{\emptyset, \{a\}\}$

b) $P(\{a,b\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$

c) $P(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

(2.1 25) a) $|P(\{a, b, \{a,b\}\})| = 2^3 = 8$

b) $|P(\{\emptyset, a, \{a\}, \{\{a\}\}\})| = 2^4 = 16$

c) $|P(P(\emptyset))| = |P(\{\emptyset\})| = 2^1 = 2$

(2.1 29) $A = \{a, b, c, d\}$ and $B = \{y, z\}$

a) $A \times B = \{(a,y), (b,y), (c,y), (d,y), (a,z), (b,z), (c,z), (d,z)\}$

b) $B \times A = \{(y,a), (y,b), (y,c), (y,d), (z,a), (z,b), (z,c), (z,d)\}$

2.1 35 a) $A=\{0,1,3\}$ $A^2=\{(0,0),(0,1),(0,3),$
$(1,0),(1,1),(1,3),$
$(3,0),(3,1),(3,3)\}$

b) $A=\{1,2,a,b\}$ $A^2=\{(1,1),(1,2),(1,a),(1,b),$
$(2,1),(2,2),(2,a),(2,b),$
$(a,1),(a,2),(a,a),(a,b),$
$(b,1),(b,2),(b,a),(b,b)\}$

2.1 37 $|A|=m$ $|B|=n$ $|A\times B|=m\cdot n$

2.1 44 Proof: Suppose A, B, C are nonempty sets and $A\times B=A\times C$. We wish to show $B=C$.

Let $x\in B$. Then $(a,x)\in A\times B$ for all $a\in A$.
Since $A\times B=A\times C$, $(a,x)\in A\times C$. Thus, $x\in C$, so $B\subseteq C$

Let $y\in C$. Then $(a,y)\in A\times C$ for all $a\in A$.
Since $A\times B=A\times C$, $(a,y)\in A\times B$. Thus $y\in B$, so $C\subseteq B$
Thus, $B=C$.

2.1 47 a) $P(x): x^2<3$ $\{-1,0,1\}$

b) $Q(x): x^2>x$ $\{\dots -3,-2,-1,2,3,\dots\}=\mathbb{Z}-\{0,1\}$

c) $R(x): 2x+1=0$ $\emptyset$ ($x=\frac{1}{2}$ is the only solution, but $\frac{1}{2}\notin\mathbb{Z}$)

2.1 48 a) $P(x): x^3\geq 1$ $\{1,2,3,\dots\}$

b) $Q(x): x^2=2$ $\emptyset$ ($x=\sqrt{2}$ is the only solution, but $\sqrt{2}\notin\mathbb{Z}$)

c) $R(x): x<x^2$ $\{\dots -3,-2,-1,2,3,\dots\}=\mathbb{Z}-\{0,1\}$

(2.2 3) $A = \{1, 2, 3, 4, 5\}$ $B = \{0, 3, 6\}$

a) $A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$

b) $A \cap B = \{3\}$

c) $A - B = \{1, 2, 4, 5\}$

d) $B - A = \{0, 6\}$

(2.2 7) a) $A \cup \mathcal{U} = \{x \mid x \in A \vee x \in \mathcal{U}\}$

$= \{x \mid x \in A \vee T\}$

$= \{x \mid T\}$

$= \mathcal{U}$

b) $A \cap \emptyset = \{x \mid x \in A \wedge x \in \emptyset\}$

$= \{x \mid x \in A \wedge F\}$

$= \{x \mid F\}$

$= \emptyset$

(2.2 9) a) $A \cup \bar{A} = \{x \mid x \in A \vee x \in \bar{A}\}$

$= \{x \mid x \in A \vee x \notin A\}$

$= \{x \mid T\}$

$= \mathcal{U}$

b) $A \cap \bar{A} = \{x \mid x \in A \wedge x \in \bar{A}\}$

$= \{x \mid x \in A \wedge x \notin A\}$

$= \{x \mid F\}$

$= \emptyset$

(2.2 25) $A \cup (B \cap C) = \{x \mid x \in A \vee x \in B \cap C\}$

$= \{x \mid x \in A \vee (x \in B \wedge x \in C)\}$

$= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\}$

$= \{x \mid x \in A \vee x \in B\} \cap \{x \mid x \in A \vee x \in C)\}$

$= (A \cup B) \cap (A \cup C)$

(2.2 27) $A = \{0,2,4,6,8,10\}$ $B = \{0,1,2,3,4,5,6\}$ $C = \{4,5,6,7,8,9,10\}$

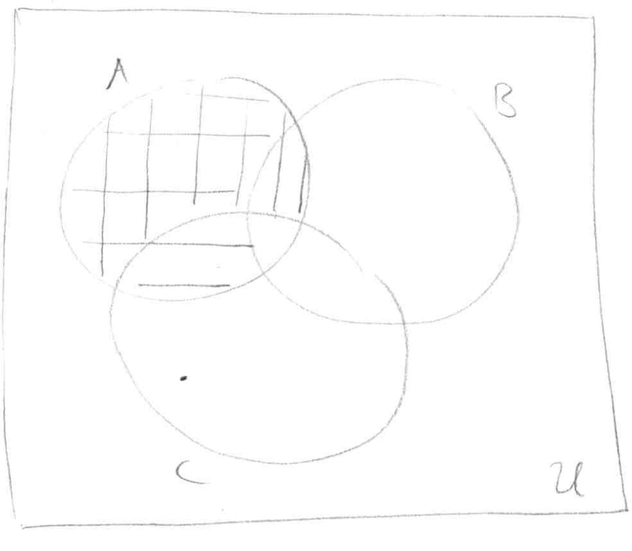
a) $A \cap B \cap C = \{4,6\}$

b) $A \cup B \cup C = \{0,1,2,3,4,5,6,7,8,9,10\}$

c) $(A \cup B) \cap C = \{4,5,6,8,10\}$

d) $(A \cap B) \cup C = \{0,2,4,5,6,7,8,9,10\}$

(2.2 29c)



(2.2 31) a) $A \cup B = A \longrightarrow B \subseteq A$

b) $A \cap B = A \longrightarrow A \subseteq B$

c) $A - B = A \longrightarrow A \cap B = \emptyset$

d) $A \cap B = B \cap A \longrightarrow$ nothing, this is always true

e) $A - B = B - A \longrightarrow A = B$

2.2 39 The symmetric difference of computer science majors and mathematics majors is the set of all students majoring in computer science, or majoring in math, but not both.

2.2 43d $A \oplus \bar{A} = \{ x \mid x \in A \oplus x \in \bar{A} \}$

$= \{ x \mid x \in A \oplus x \notin A \}$

$= \{ x \mid T \}$

$= \mathcal{U}$.

2.2 47 Proof: Suppose that A, B, C are sets such that $A \oplus C = B \oplus C$. We wish to show that $A = B$. We first show that $A \subseteq B$.

Let $x \in A$. Now we have 2 cases: either $x \in C$ or $x \notin C$.

Case 1: Suppose $x \in C$.

Since $x \in A$ and $x \in C$, $x \notin A \oplus C$.

Since $A \oplus C = B \oplus C$, $x \notin B \oplus C$.

Since $x \notin B \oplus C$, x must be in both B and C or neither.

However, since we assumed $x \in C$, x can't be in neither.

Thus, x is in both B and C, so x is in B.

Case 2: Suppose $x \notin C$.

Since $x \in A$ and $x \notin C$, $x \in A \oplus C$.

Since $A \oplus C = B \oplus C$, $x \in B \oplus C$.

Since $x \in B \oplus C$, x is in B or C but not both.

However, since we assumed $x \notin C$, it must be the case that $x \in B$.

To show $B \subseteq A$, we can repeat the same steps as above

(this is a "without loss of generality" scenario"

Ξ ©

2.2 49

A	B	C	D	$A \oplus B$	$C \oplus D$	$(A \oplus B) \oplus (C \oplus D)$	$A \oplus C$	$B \oplus D$	$(A \oplus C) \oplus (B \oplus D)$
1	1	1	1	0	0	0	0	0	0
1	1	1	0	0	1	1	0	1	1
1	1	0	1	0	1	1	1	0	1
1	1	0	0	0	0	0	1	1	0
1	0	1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0	0	0
1	0	0	1	1	1	0	1	1	0
1	0	0	0	1	0	1	1	0	1
0	1	1	1	1	0	1	1	0	1
0	1	1	0	1	1	0	1	1	0
0	1	0	1	1	1	0	0	0	0
0	1	0	0	1	0	1	0	1	1
0	0	1	1	0	0	0	1	1	0
0	0	1	0	0	1	1	1	0	1
0	0	0	1	0	1	1	0	1	1
0	0	0	0	0	0	0	0	0	0

2.2 50

Proof: Let A and B be finite sets.

Suppose $|A| = m$ and $|B| = n$.

Then $|A \cup B| \leq m+n$

Thus, $A \cup B$ is finite. ∎

2.2 54

Let $A_i = \{\ldots, -2, -1, 0, 1, \ldots, i\}$

a) $\bigcup_{i=1}^{n} A_i = \{\ldots, -2, -1, 0, 1, \ldots, n\}$

b) $\bigcap_{i=1}^{n} A_i = \{\ldots, -2, -1, 0, 1\}$

2.2
56

a) $A_i = \{ i, i+1, i+2, \dots \}$

$$\bigcup_{i=1}^{\infty} A_i = \{1, 2, 3, \dots\} = \mathbb{Z}^+$$

$$\bigcap_{i=1}^{\infty} A_i = \emptyset$$

b) $A_i = \{0, i\}$

$$\bigcup_{i=1}^{\infty} A_i = \{0, 1, 2, 3, \dots\} = \mathbb{N}$$

$$\bigcap_{i=1}^{\infty} A_i = \{0\}$$

c) $A_i = (0, i)$

$$\bigcup_{i=1}^{\infty} A_i = (0, \infty) = \mathbb{R}^+$$

$$\bigcap_{i=1}^{\infty} A_i = (0, 1)$$

d) $A_i = (i, \infty)$

$$\bigcup_{i=1}^{\infty} A_i = (1, \infty)$$

$$\bigcap_{i=1}^{\infty} A_i = \emptyset$$

2.2
59

a) 11 1100 1111 $\{1, 2, 3, 4, 7, 8, 9, 10\}$

b) 01 0111 1000 $\{2, 4, 5, 6, 7\}$

c) 10 0000 0001 $\{1, 10\}$

2.2
67

$A = \{3 a, 2 \cdot b, 1 \cdot c\}$ $B = \{2 \cdot a, 3 \cdot b, 4 \cdot d\}$

a) $A \cup B = \{3 \cdot a, 3 \cdot b, 1 \cdot c, 4 \cdot d\}$

b) $A \cap B = \{2 \cdot a, 2 \cdot b\}$

c) $A - B = \{1 \cdot a, 1 \cdot c\}$

d) $B - A = \{1 \cdot b, 4 \cdot d\}$

e) $A + B = \{5 \cdot a, 5 \cdot b, 1 \cdot c, 4 \cdot d\}$